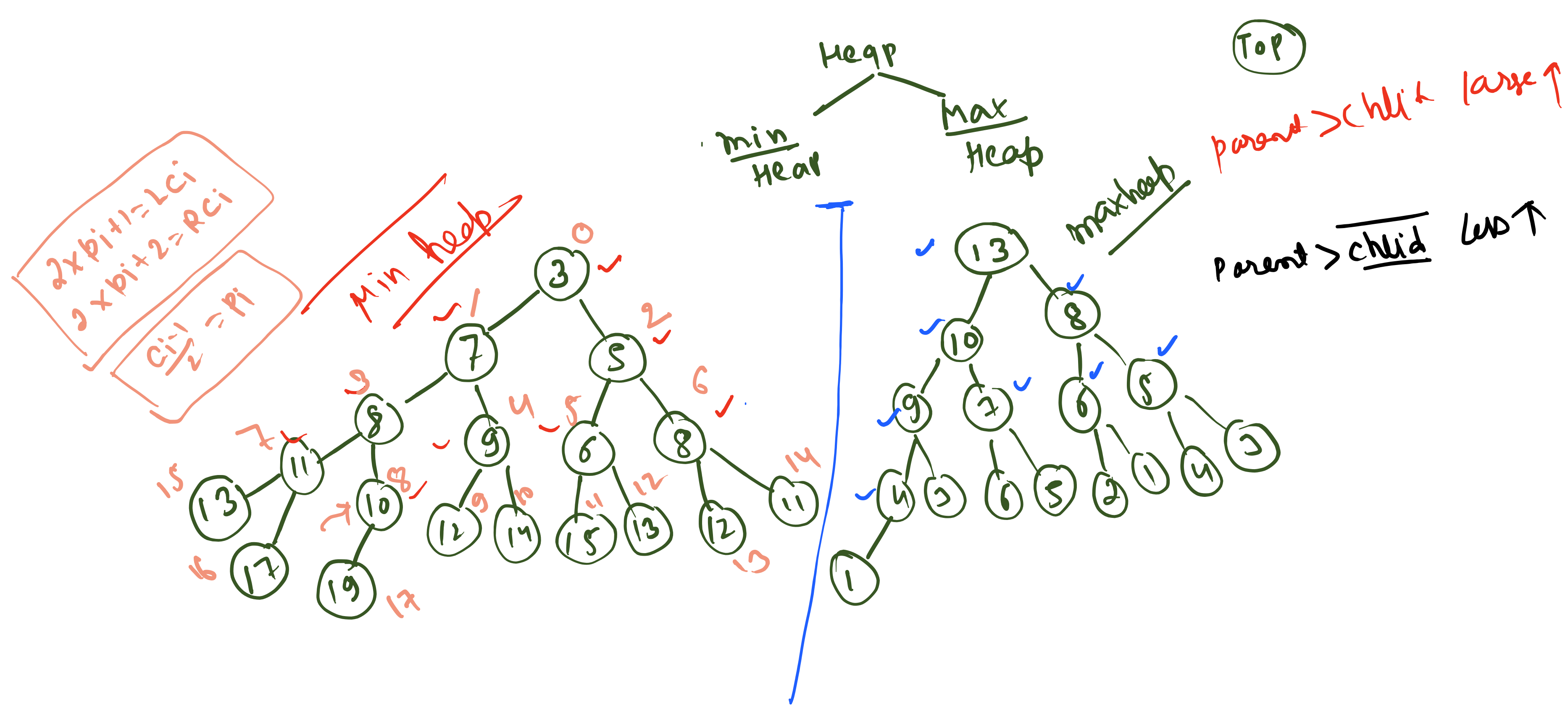
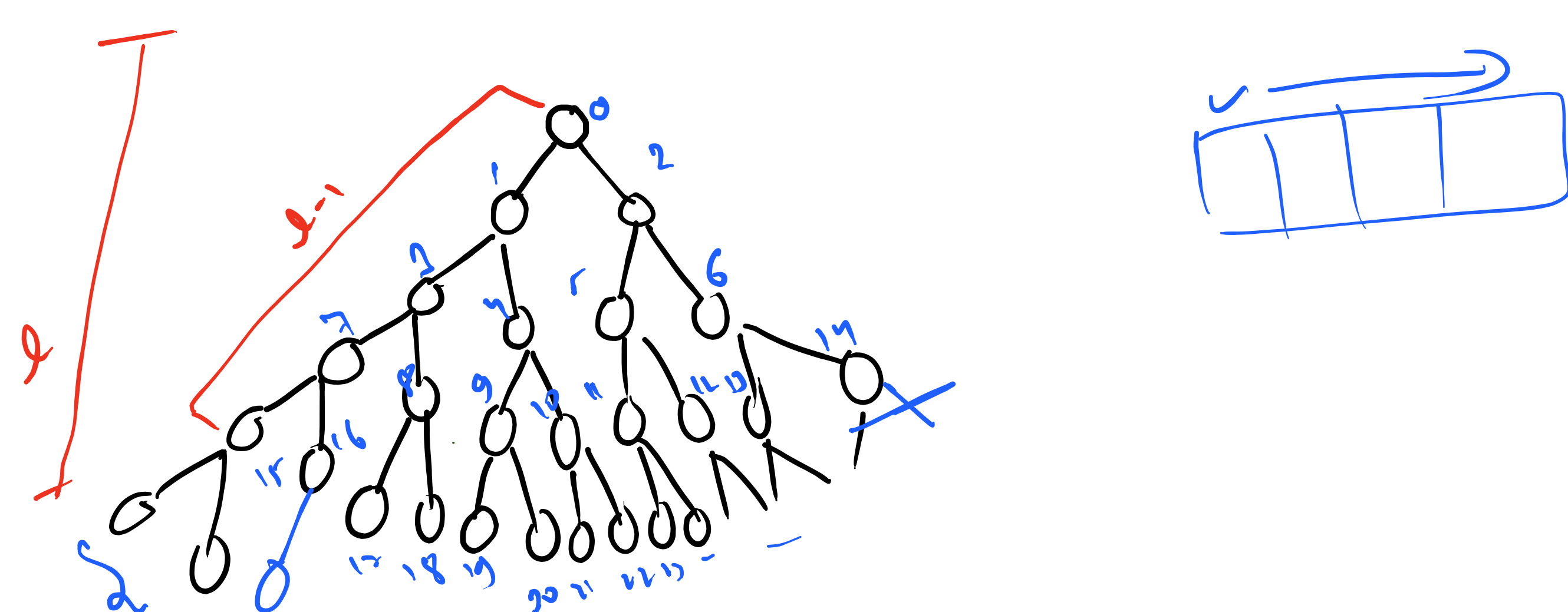
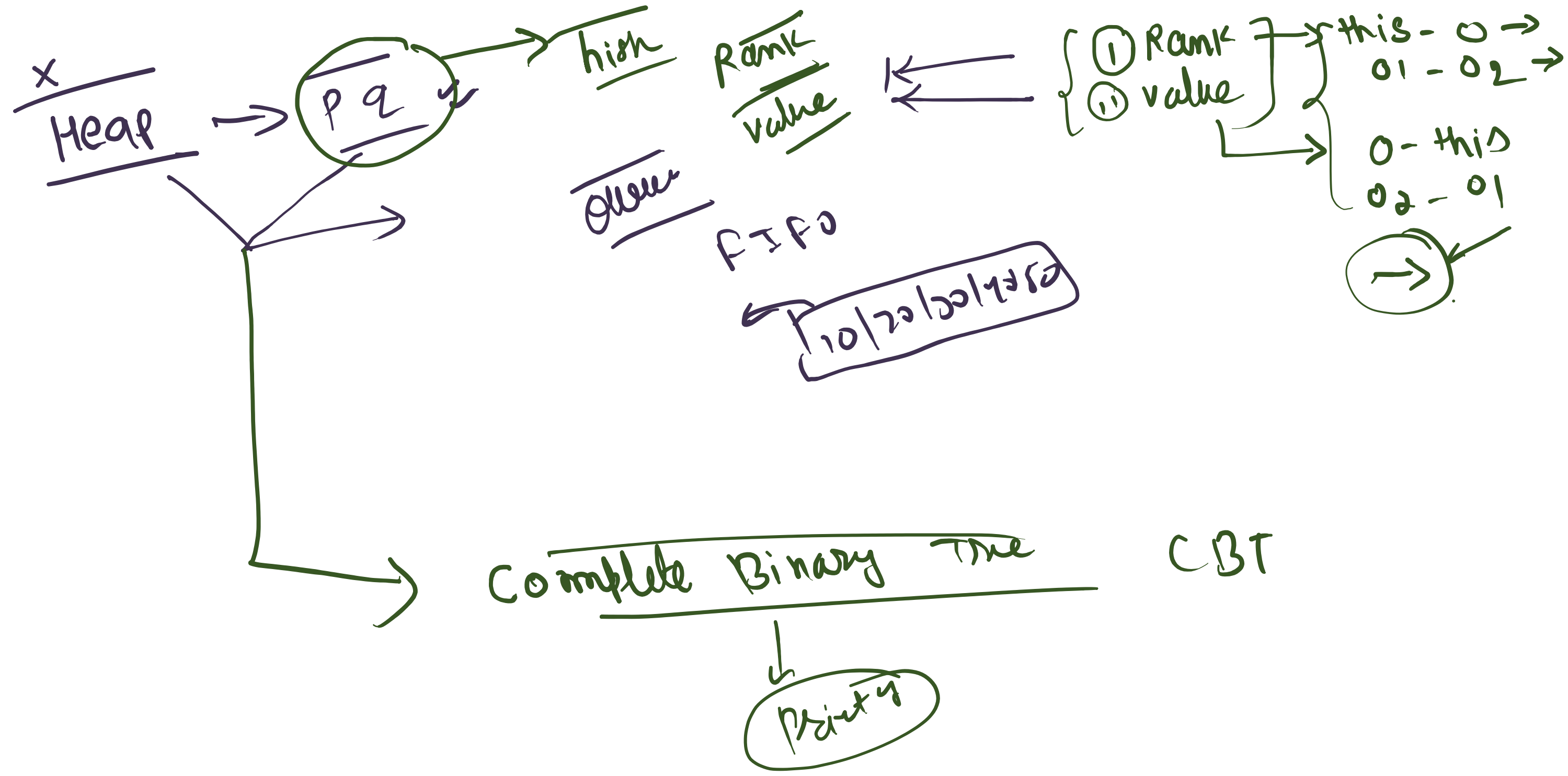
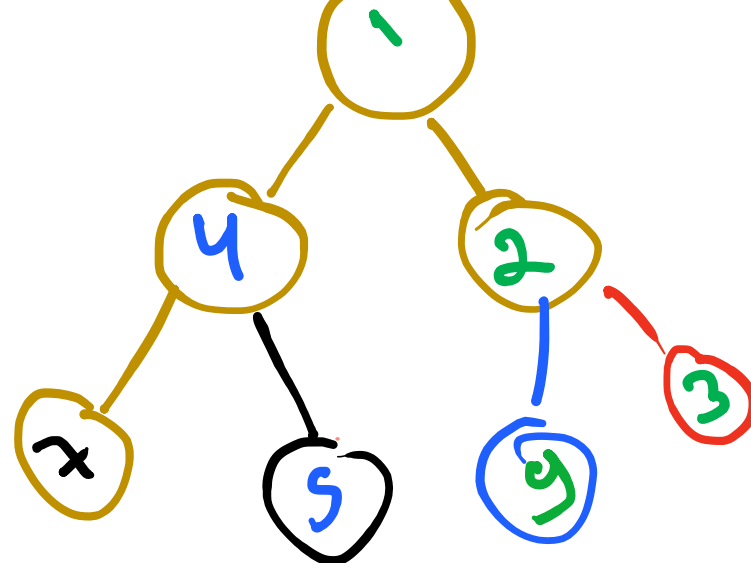




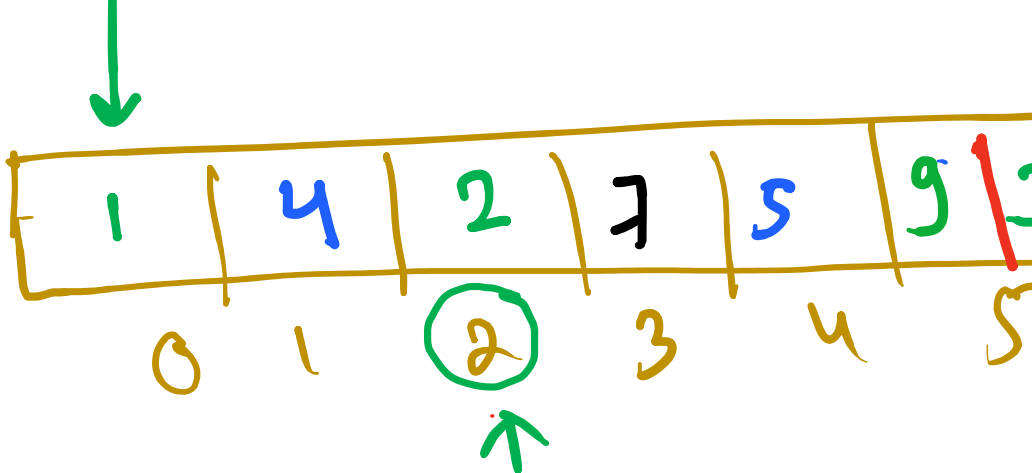
	Sorted data	Unsorted data	Heap
add	$O(n)$	$O(n)$	$\log(n) \rightarrow ?$
min remove	$O(n)$	$O(n)$	$\log(n) \rightarrow ?$
min get	$O(1)$	$O(n)$	$O(1)$

$3^{th} = 3$

[0, 1, 2, 3, 4, 5]



```
private void upheapify(int ci) {  
    // TODO Auto-generated method stub  
    int pi = (ci - 1) / 2;  
    if (list.get(pi) > list.get(ci)) {  
        swap(pi, ci);  
    }  
}
```

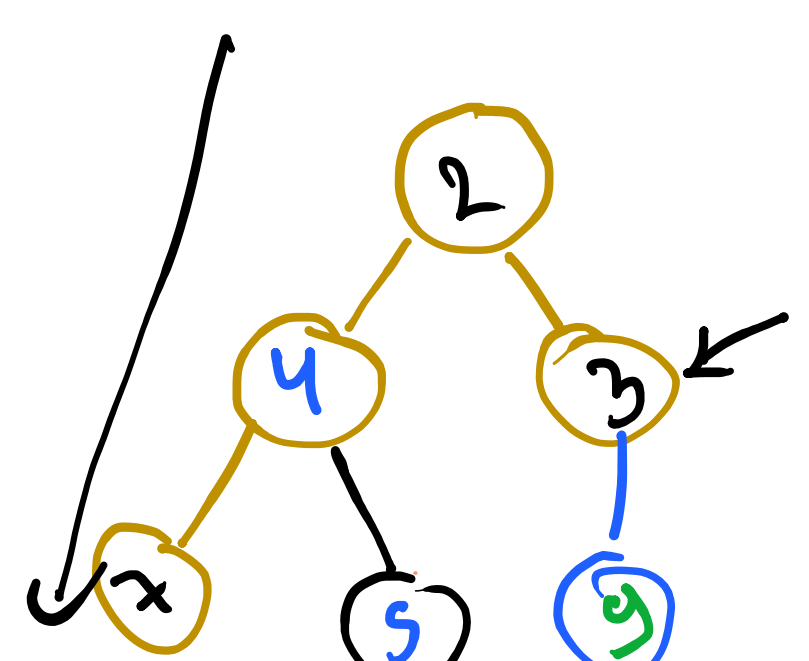


$\frac{(1+6+10)}{2}$

$\frac{1+6}{2} = 3.5$
 $\frac{1+10}{2} = 5.5$
 $\frac{3.5+5.5}{2} = 4.5$
 $\frac{4.5+9}{2} = 6.75$
 $\frac{6.75+13}{2} = 9.875$

[0, 1, 2, 3, 4, 5, 6]

[2, 4, 3, 7, 5, 9, 1]



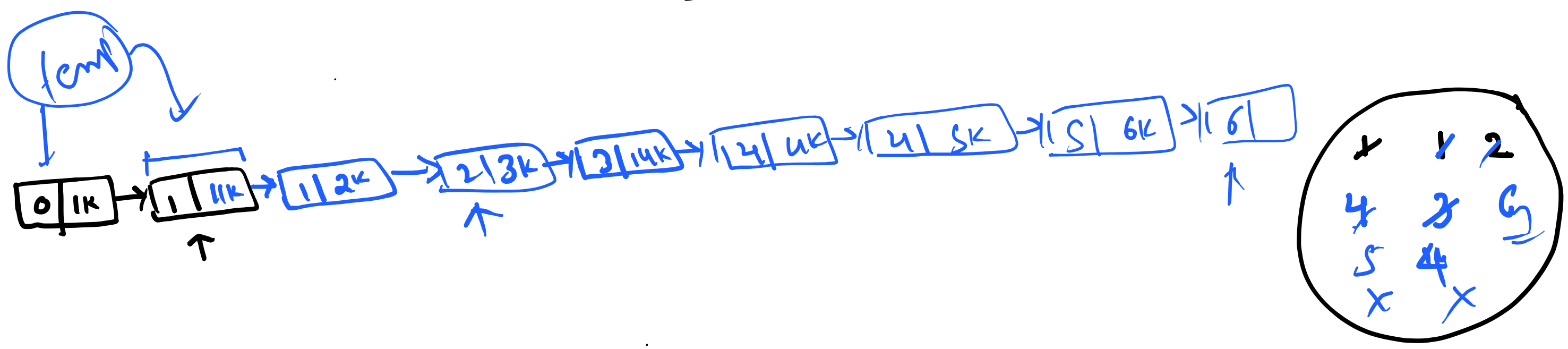
```
private void downheapify(int pi) {  
    // TODO Auto-generated method stub  
    int lci = 2 * pi + 1;  
    int rci = 2 * pi + 2;  
    int mini = pi;  
    if (list.get(lci) < list.get(mini)) {  
        mini = lci;  
    }  
    if (list.get(rci) < list.get(mini)) {  
        mini = rci;  
    }  
    if (mini != pi) {  
        swap(pi, mini);  
        downheapify(mini);  
    }  
}
```

[3, 2, 3, 1, 2, 4, 5, 5, 6, 1, 2, 3], k = 4

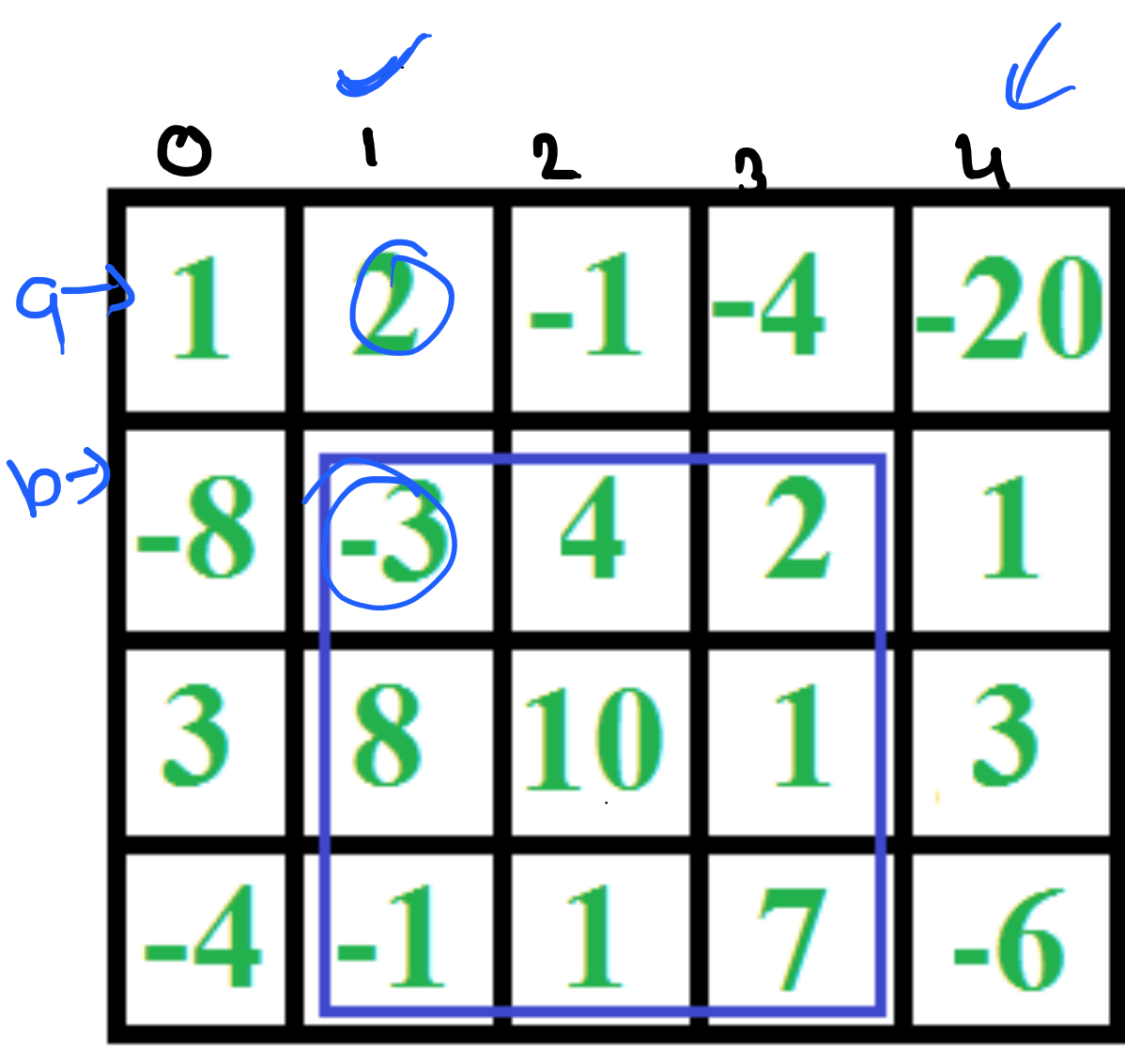
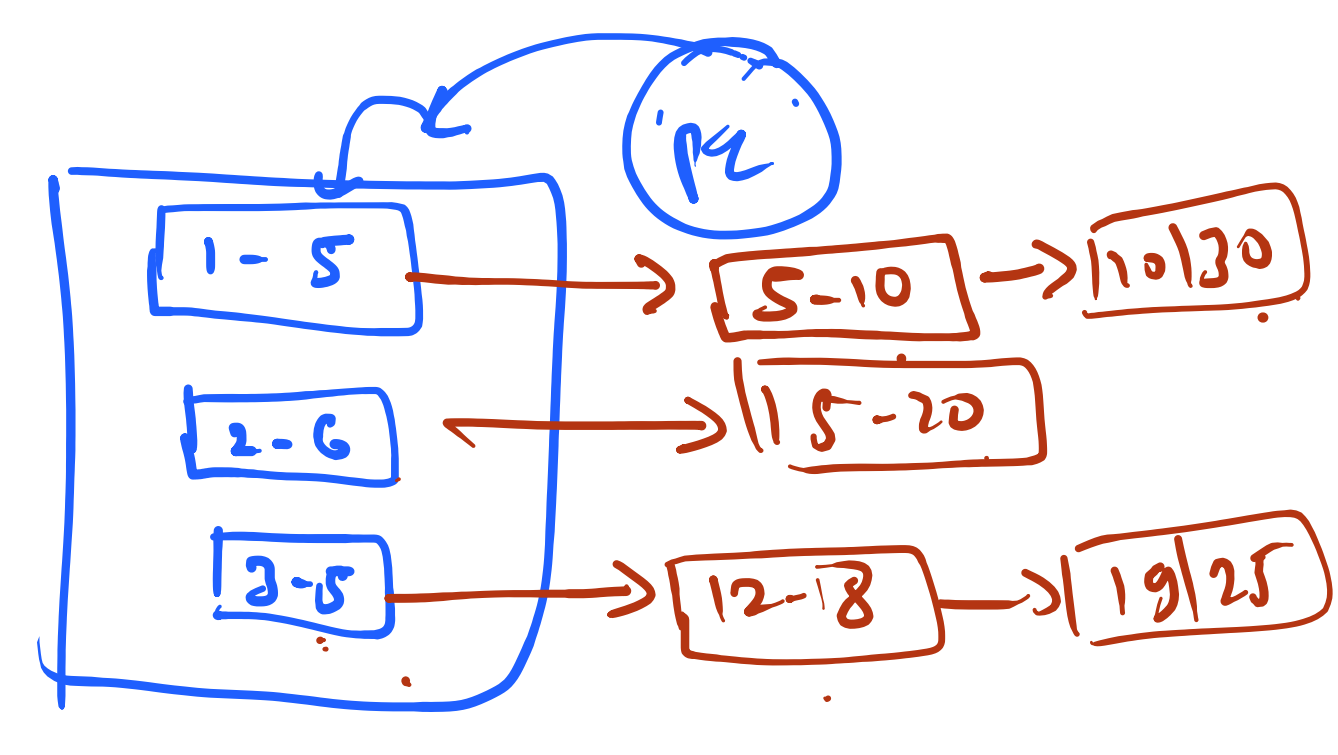
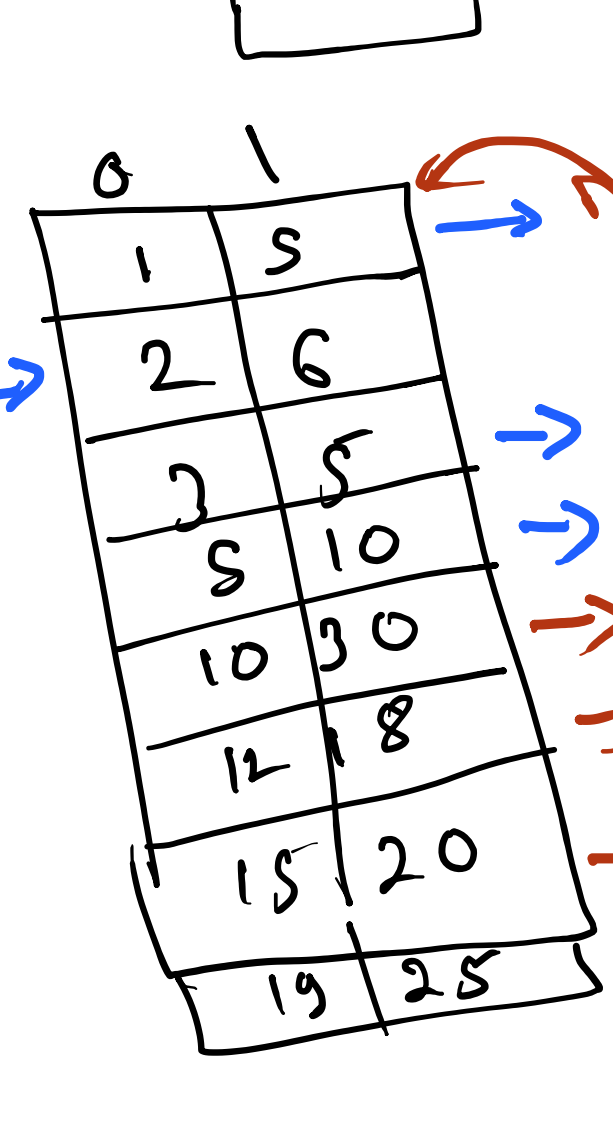
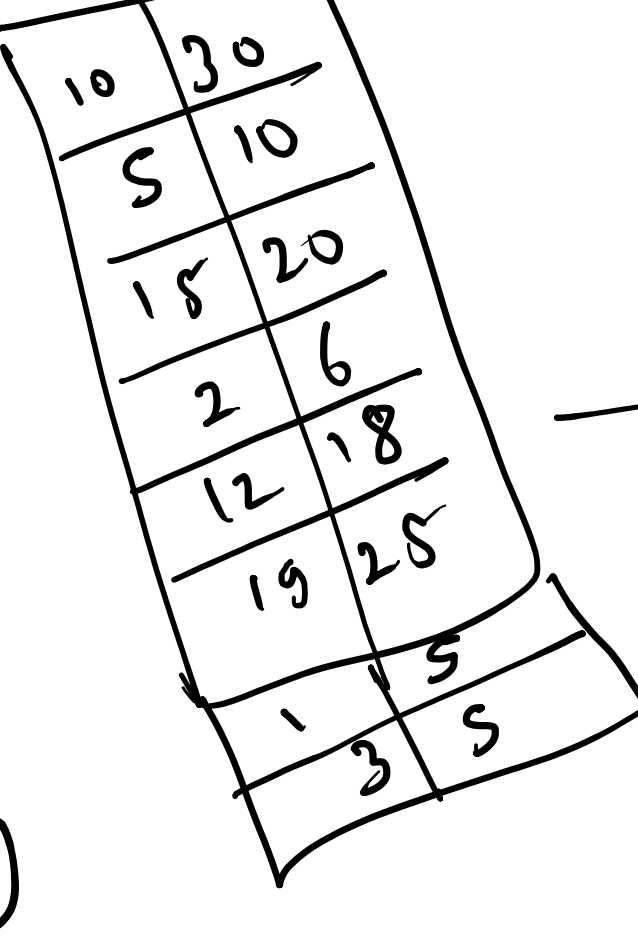
5564

Input: Lists = [[1,4,5],[1,3,4],[2,6]]

[1, 4, 5] [1, 3, 4] [2, 6]

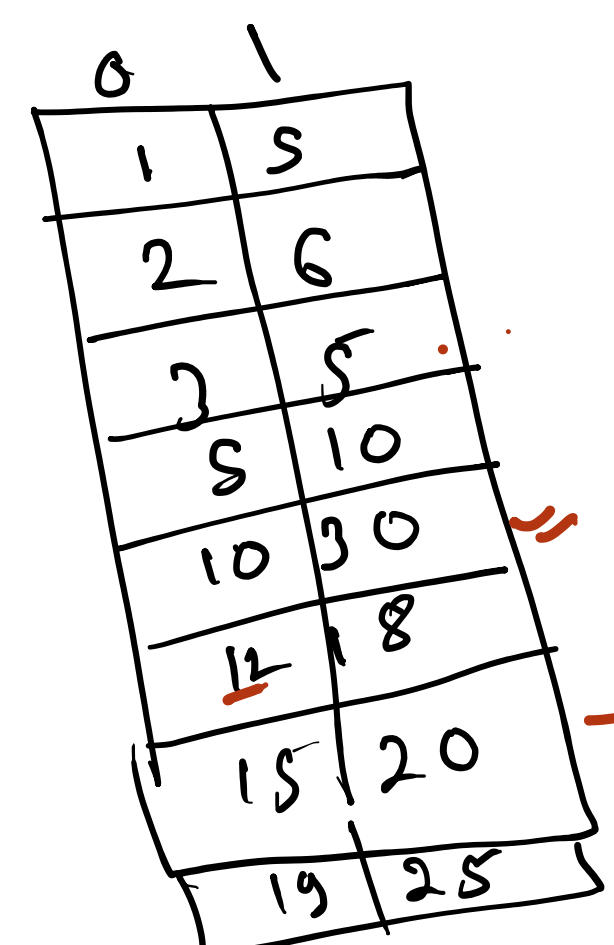


[10, 20]
(5, 10)
(15, 20)
(2, 6)
(12, 18)
(19, 25)
(1, 5)
(3, 15)



-8 -3 4 2 1
-4 -1 7 6
1 2 -4 -20
3 8 10 1 3
2 2 -1 -4 -20
-4 -1 7 -6
-8 -3 4 2 1
3 8 10 1 3

```
public static int Meeting_Rooms(int[][] intervals) {  
    Arrays.sort(intervals, (a, b) -> a[0] - b[0]);  
    PriorityQueue<int> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]);  
    pq.add(intervals[0]);  
    for (int i = 1; i < intervals.length; i++) {  
        if (intervals[i][0] >= pq.peek()[1]) {  
            pq.poll();  
        }  
        pq.add(intervals[i]);  
    }  
    return pq.size();  
}
```



[1, 5] -> [5, 10]
[2, 6] -> [12, 18]
[3, 8] -> [19, 25]